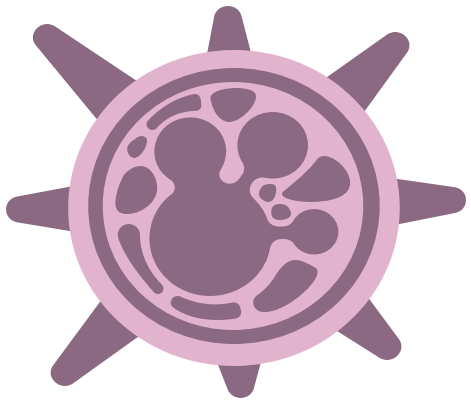
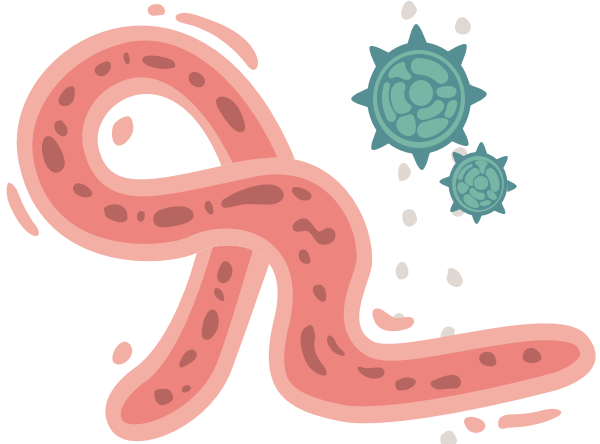


Comparative Analysis of TEs in *Trichinella spiralis* and T8

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Introduction

WHAT IS TRICHINELLA?

- Trichinella is a genus of parasite nematode that causes trichinellosis, a food-borne disease (Zarlenga et al., 2020).
- Transmission occurs through the consumption of raw or undercooked infected meat (Zarlenga et al., 2020).
- The parasite has a worldwide distribution and infects a range of hosts (Zarlenga et al., 2020).

T. spiralis

- *T. spiralis* (T1) is a widely distributed, encapsulated Trichinella species commonly associated with domestic pigs and rodents (Zarlenga et al., 2020).
- Unlike T8, *T. spiralis* can be maintained in domestic transmission cycles and has historically been a major cause of human trichinellosis through consumption of infected pork (Zarlenga et al., 2009).

T8 Genotype

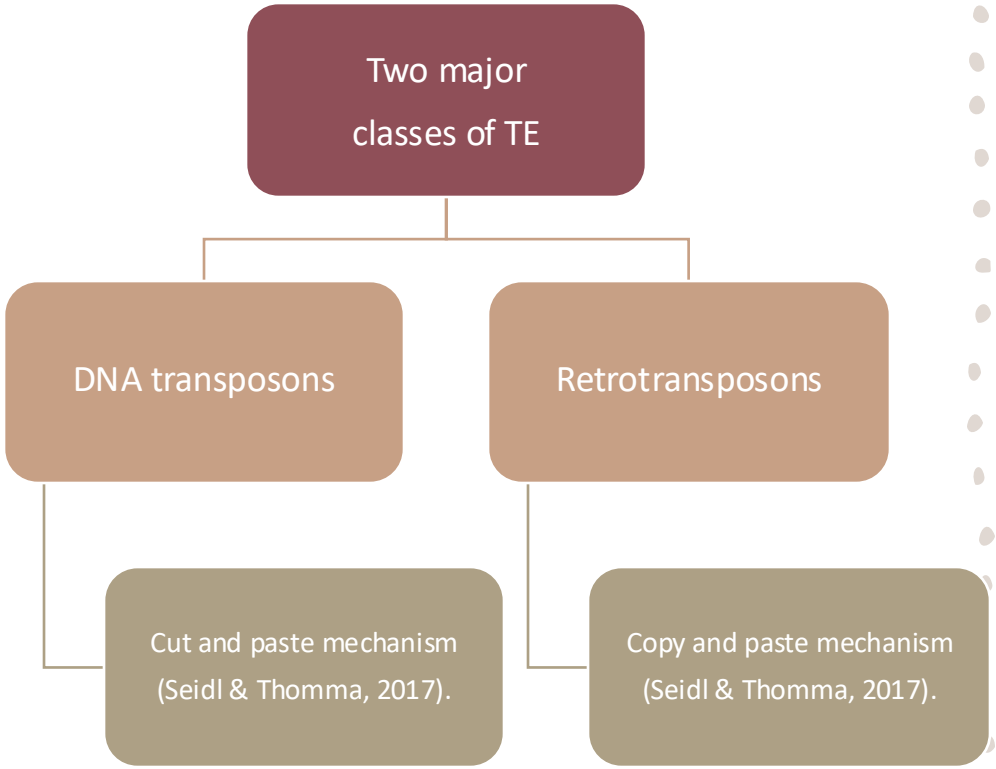
- T8 is an African encapsulated Trichinella genotype that diverged from *T. Britovi* 3.2-1.7 million years ago (Korhonen et al., 2016).
- Geographic isolation and environmental change promoted diversification of T8 (Korhonen et al., 2016).

Comparison

Characteristics	*Trichinella spiralis*	*Trichinella T8*
Taxonomic status	Valid species	Genotype
Geographic range	Worldwide	Southern Africa (primarily Naimbia and South Africa)
Domestic cycle	Commonly established in domestic pigs	Unknown
Primary hosts	Domestic and wild mammals	Primarily wild carnivores
Human Infection	Major cause of human trichinellosis worldwide	No confirmed significant role in human disease
Evolutionary adaptation	Domestic and sylvatic environments	Sylvatic (wildlife) environments
Research importance	Most extensively studied species	Poorly studied genotype

Transposable elements (TE)

- TE are DNA sequences that can move to new positions within a genome (Seidl & Thomma, 2017).
- They generate genetic diversity by disrupting genes through insertion, causing chromosomal rearrangements, altering expression of nearby genes, and creating new regulatory elements (Seidl & Thomma, 2017).
- Most TE insertions are neutral or deleterious and eliminated by natural selection, but they also be beneficial. (Seidl & Thomma, 2017).



Research Background

- *Trichinella spiralis* and *trichinella* T8 are closely related parasitic nematodes with similar life cycles but differ in host associations, adaptation patterns, and demographics.
- This difference in host associations provides as interesting basis for investigating whether their TE classes exhibit different patterns.
- Retrotransposons are mobile genetic elements that replicate through copy-and-paste mechanism, allowing for better accumulation throughout the genome. Their activity contributes to genetic variation by influencing gene structure and regulation, making them useful in exploring evolutionary differences between organisms (Wells & Feschotte, 2020; Jangam et al., 2017) .

Research Question

Do differences in retrotransposons insertion and deletion patterns between *T. spiralis* and T8 reveal patterns in how each organism adapt to their host?

- To carry out this study, a comparative analysis of retrotransposon insertion and deletion patterns in both organisms was conducted, examining their overall distributions and differences across individual retrotransposon classes.

Methods

Reading CSV files:

```
trich_main = read_csv("../data/GCA_001447595.1.csv")

trich_t8 = read_csv("../data/GCA_001447745.1.csv")
```

Filtering for Retro transposons in both species:

```
trich_main_retro = trich_main |> filter(str_detect(matching_class, "^(LTR|PLE|SINE|LINE)"))

trich_t8_retro = trich_t8 |>  filter(str_detect(matching_class, "^(LTR|LINE|SINE|PLE)"))
```

Boxplots for % deletion for both species:

```
trich_main_retro |> ggplot(aes(x = matching_class, y = perc_del)) + geom_boxplot() +

labs(title = "T. spiralis Retrotransposons", y ="Percent Deletions (%)" , x = "Retrotransposons class"
```

```
trich_t8_retro |> ggplot(aes(x = matching_class, y = perc_del)) + geom_boxplot() +

labs(title = "Trich T8 Retrotransposons", y ="Percent Deletions (%)" , x = "Retrotransposons class")
```

Boxplots for % insertions for both species:

```
trich_main_retro |> ggplot(aes(x = matching_class, y = perc_insert)) + geom_boxplot() +

labs(title = "T. spiralis Retrotransposons", y ="Percent Insertions (%)" ,

x = "Retrotransposons class")

trich_t8_retro |> ggplot(aes(x = matching_class, y = perc_insert)) + geom_boxplot() +

labs(title = "Trich T8 Retrotransposons", y ="Percent Insertions (%)" ,

x ="Retrotransposons class")
```

Combining retrotransposons insertion/deletion between both species:

```
retro_combined = bind_rows(trich_main_retro |>

mutate(species = "T. spiralis"), trich_t8_retro |>

mutate(species = "T8"))
```

Boxplot for % insertion/deletion for each specie:

```
retro_combined |> ggplot(aes(x = matching_class, y = perc_del, fill = species)) + geom_boxplot() +

facet_wrap(~species) + labs( title = "Deletion Percentages Across Retrotransposon Classes",

x = "Retrotransposon Class",

y = "Percentage Deletions (%)" ) + theme_bw() + theme(axis.text.x = element_text(angle = 45, hjust = 1))

retro_combined |> ggplot(aes(x = matching_class, y = perc_insert, fill = species)) + geom_boxplot() +

facet_wrap(~species) + labs(title = "Insertion Percentages Across Retrotransposon Classes", x

 = "Retrotransposon Class",

y = "Percentage Insertions (%)" )  +  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Retrotransposon Composition:

```
retro_combined |> count(species, matching_class) |> group_by(species) |> mutate(percent = n / sum(n) * 100) |> ggplot(aes(x = species, y = percent, fill = matching_class)) + geom_col() + labs (title = "Retrotransposon Composition", x = "Genome", y = "Percentage of Annotations", fill = "Retrotransposon Class") + theme_minimal()
```

Percent-Difference Comparison:

```
retro_combined |>  count(species, matching_class) |> group_by(species) |> mutate(percent = n / sum(n) * 100) |> select(species, matching_class, percent) |> pivot_wider (names_from = species, values_from = percent, values_fill = 0) |> mutate(difference = T8 - `T. spiralis`) |> arrange(desc(abs(difference)))
```

Statistical Significance (χ^2test):

```
class_counts <- retro_combined |> count(matching_class, species) |> pivot_wider(names_from = species, values_from = n, values_fill = 0)

count_matrix <- class_counts |> select(`T. spiralis`, T8) |> as.matrix()

chi_result <- chisq.test(count_matrix)
```

Figure 1

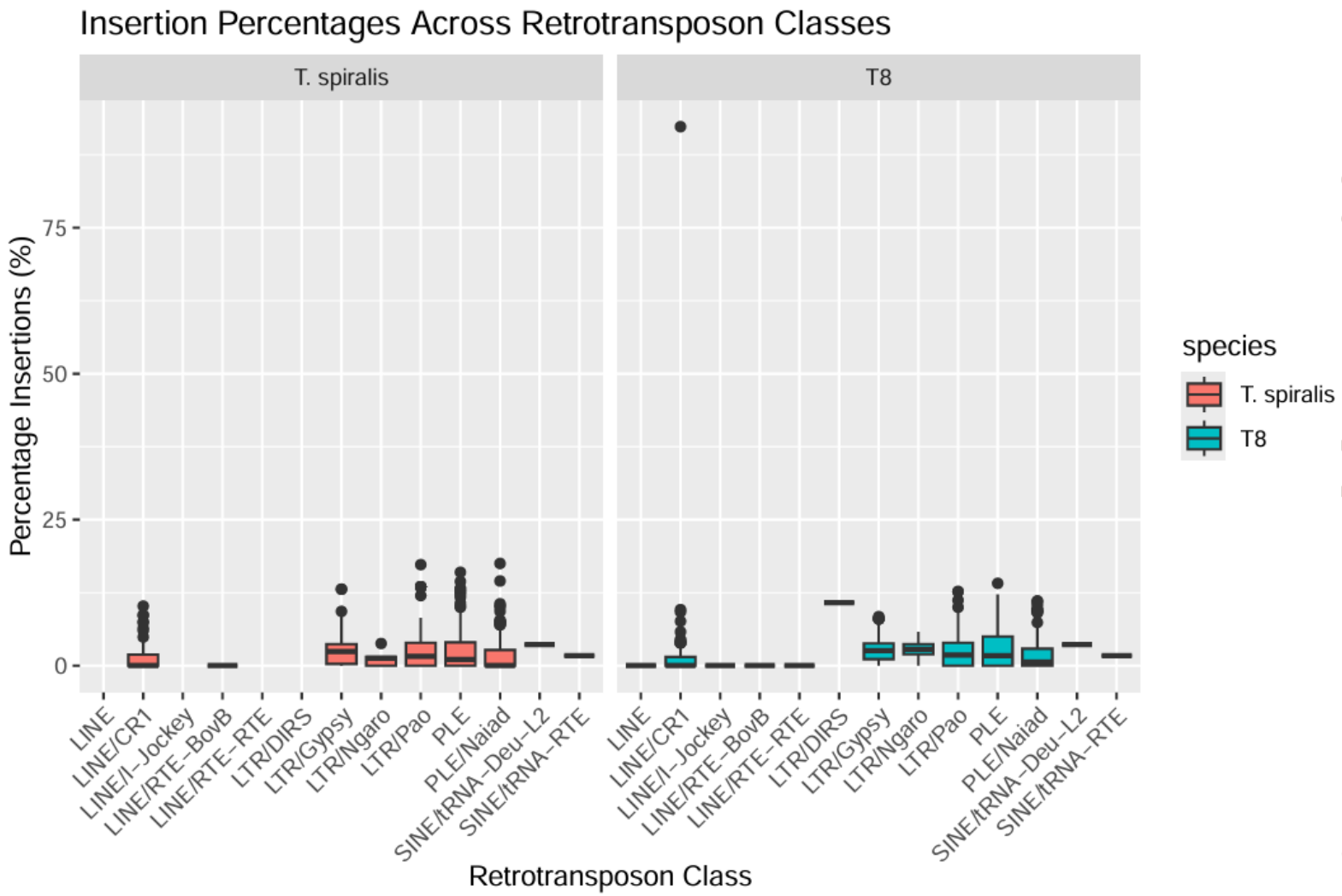


Figure 1: Comparison of insertion percentages across retrotransposon classes in *Trichinella spiralis* and *Trichinella* sp. T8.

Figure 2

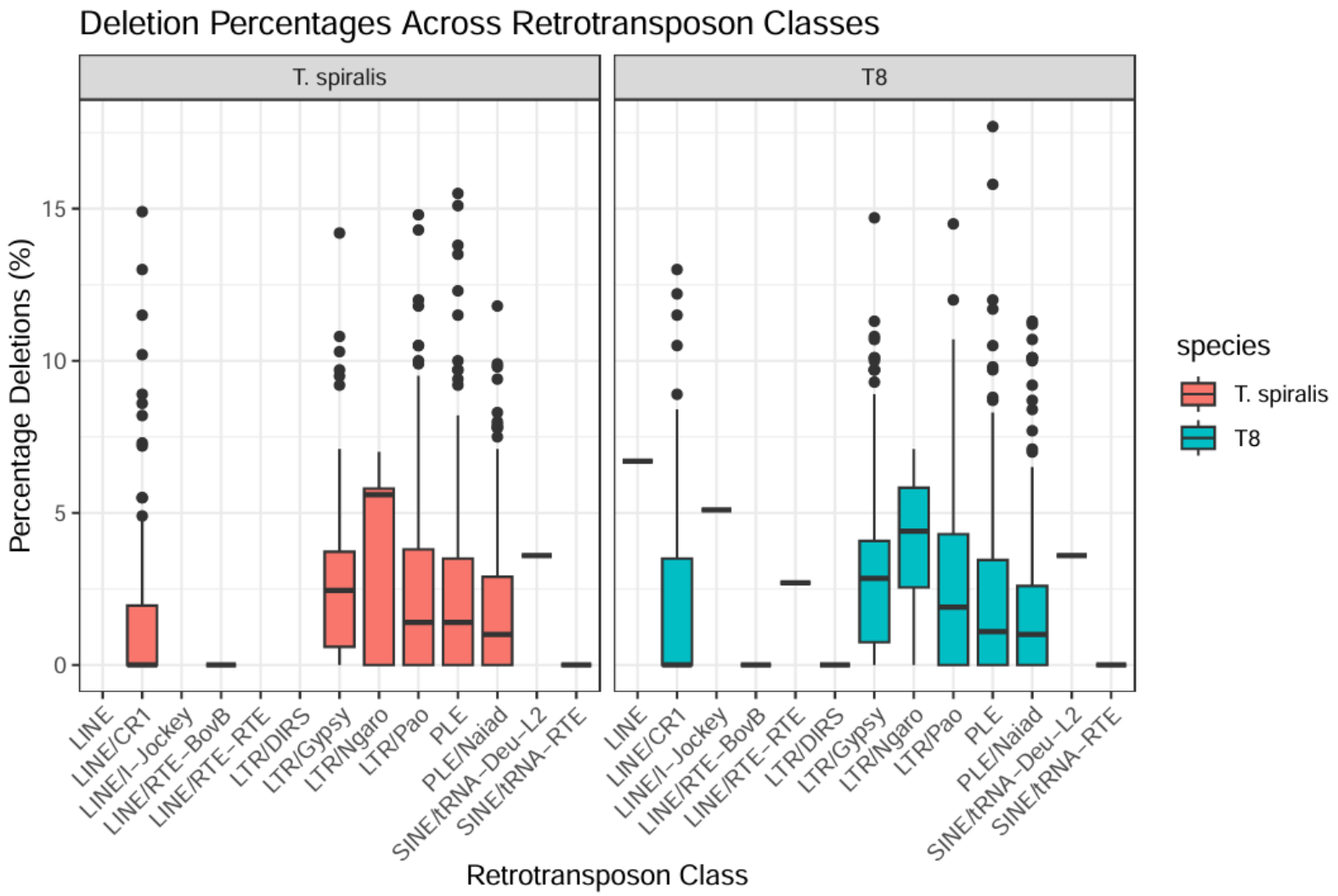


Figure 2: Comparison of deletion percentages across retrotransposon classes in *Trichinella spiralis* and *Trichinella* sp. T8.

Figure 3

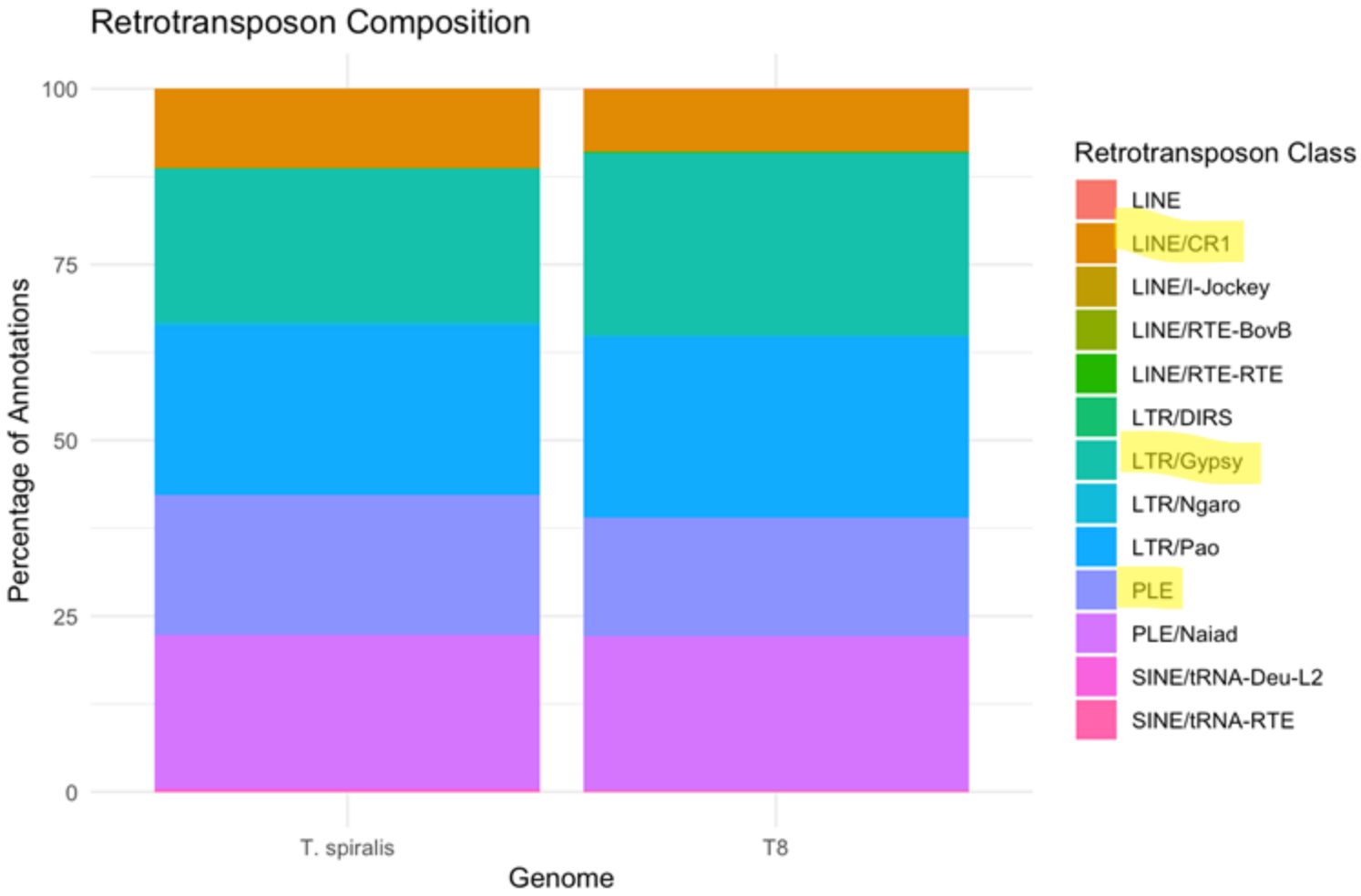


Figure 3: Relative distribution of retrotransposon classes in *Trichinella spiralis* and *Trichinella sp*. T8. Stacked bars represent the percentage of retrotransposon annotations from each class within each genome.

Results and Discussion

Retrotransposon insertion – Figure 1

- *T. spiralis* and T8 exhibit broadly similar insertion percentage distributions across most retrotransposon classes.
- Most classes show low median insertion percentages, with some outliers
- Relevance to host adaptation: Similarity in overall insertion patterns does not rule out differences in genes that may influence how each species adapts to its host.

KEY TAKEAWAY: The similar insertion patterns between *T. spiralis* and T8 do not show a clear connection to how these organisms adapt to their hosts.

Results and Discussion

Retrotransposon deletion – Figure 2

- Both organisms exhibit overlapping deletion distributions across most retrotransposon classes
- Some classes specific differences are observed, particularly LINE, LINE/Jockey, and LTR/Ngaro.
- Relevance to host adaptation: Differences in deletion patterns may affect genome structure and gene regulation, potentially influencing host adaptation.

KEY TAKEAWAY: Similar to insertion patterns, deletion patterns alone cannot establish a relationship between retrotransposons and host adaptation.

Results and Discussion

Retrotransposon composition – Figure 3

- LTR/Gypsy: +4% in T8, PLE: -3%, LINE/CR1 -2.5%, X^2= 13.398, df = 12, p-value = 0.340
- The similarity in retrotransposon composition indicates that host adaptation cannot be attributed to variation in TE composition among *T. spiralis* genotypes alone.

- The low frequency of some retrotransposon classes means the chi-squared test is not fully reliable.

What next?

- We found that despite T. spiralis and T8 having very different host-parasite interactions, their overall retrotransposon composition is similar. If TE composition is not solely responsible for host adaptation, then future research could investigate whether TE location has any bearing on host interactions.
- **Future Question**: "Do T. Spiralis and T8 differ in TEs located near genes associated with host interaction?"

Novelty and Creativity

- Expanded the analysis to examine the relative proportions of retrotransposon classes between the two genomes (Fig. 3)
- Converted raw annotation counts into percentages to allow fairer comparisons of retrotransposon class composition between genomes (relative abundance comparison after normalization).
- Identified the retrotransposon class (LTR/Gypsy) that had the largest difference in relative composition (4% higher proportion of retrotransposon annotations in T8).

References (APA)

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Copilot Prompts

- "Which of these classes are retrotransposons?"
- "Given the list of Trichinella species, which ones count and encapsulated and non-encapsulated. Also provide a brief overview on each of these species."